

Lecture 4: Modular Forms

Mini-Course: Magma and the LMFDB

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MIT

Lecture Overview

Modularity, the Unifying Theme

Modular Forms: The Setup

The Modularity Bridge

Research Frontier: Murmurations of Modular Forms

Closing the Loop: Theta Series Are Modular Forms

Beyond Genus 1: Genus 2 Curves

The Web of the Mini-Course

Modularity, the Unifying Theme

Where We've Been, Where We're Going

Three lectures in, here is the web we have built:

- **Lecture 1.** $\theta_{E_{16}} = \theta_{E_8 \oplus E_8}$ – two non-isometric lattices, same generating series. *What kind of object is this q -series?*
- **Lecture 2.** Number fields, Galois groups, class fields. *Where do their L -functions live?*
- **Lecture 3.** Every elliptic curve over $\mathbb{Q}$ has an attached newform. *What is this newform, concretely?*

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Today's research frontier: *murmurations of modular forms* (Zubrilina, 2024), the modular-form analog of last lecture's elliptic-curve discovery.

Modular Forms: The Setup

What are Modular Forms?

Motivation. A lattice $\Lambda \subset \mathbb{C}$ determines an elliptic curve $\mathbb{C}/\Lambda$. A function on lattices

$$F : \{\text{lattices in } \mathbb{C}\} \rightarrow \mathbb{C}, \quad F(\lambda\Lambda) = \lambda^{-k}F(\Lambda).$$

is determined by $f(z) := F(\langle 1, z \rangle)$ on $\mathcal{H}$, and the scaling relation becomes

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}). \quad (1)$$

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Definition (Modular forms)

$M_k(\Gamma_0(N))$: holomorphic $f : \mathcal{H} \rightarrow \mathbb{C}$ satisfying (1) for $\gamma \in \Gamma_0(N)$, holomorphic at the cusps (the finitely many “boundary points” of $\Gamma_0(N) \backslash \mathcal{H}$ – non-negative q -expansion at each). Its subspace of **cusp forms** $S_k(\Gamma_0(N))$ vanishes at every cusp.

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Since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma_0(N)$, f is 1-periodic: $f(z) = \sum_{n \geq 0} a_n q^n$, $q = e^{2\pi iz}$.

Spaces of Modular Forms

For fixed N and k , $M_k(\Gamma_0(N))$ is a finite-dimensional $\mathbb{C}$ -vector space:

$$M_k(\Gamma_0(N)) = S_k(\Gamma_0(N)) \oplus E_k(\Gamma_0(N)),$$

where E_k is spanned by Eisenstein series. Dimensions are given by explicit closed formulas.

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Small cases.

- $\dim S_2(\Gamma_0(11)) = 1$ – a single weight-2 newform of level 11 [11.2.a.a](#).
- $\dim S_2(\Gamma_0(37)) = 2$ – two weight-2 newforms, [37.2.a.a](#) and [37.2.a.b](#). The former is the modular form of our running curve [37.a1](#).
- $\dim S_{12}(\Gamma_0(1)) = 1$ – spanned by $\Delta(z) = q \prod_n (1 - q^n)^{24}$. Its coefficients satisfy $|\tau(p)| \leq 2p^{11/2}$ (Deligne, 1974) – the weight-12 version of $|a_p(E)| \leq 2\sqrt{p}$ from Lecture 3.
- $\dim M_4(\Gamma_0(1)) = 1$ – spanned by the Eisenstein series E_4 .

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Punchline. These spaces are small, structured, and computable. Modular forms tabulated up to weight ~ 12 and level $\sim 10\,000$ on the LMFDB.

Modular Forms in Magma

```
1 > M := ModularForms(Gamma0(37), 2);
2 > Dimension(M);
3 3 // = dim S_2 + dim E_2 = 2 + 1
4 > S := CuspidalSubspace(M);
5 > Dimension(S);
6 2
7 > NN := Newforms(S);
8 > [* qExpansion(NN[i][1], 10) : i in [1..#NN] *];
9 [*
10     q - 2*q^2 - 3*q^3 + 2*q^4 - 2*q^5 + 6*q^6 - q^7 + 6*q^9 + O(q^10),
11     q + q^3 - 2*q^4 - q^7 - 2*q^9 + O(q^10)
12 *]
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```

Recognize the first newform? $a_2 = -2, a_3 = -3, a_5 = -2, a_7 = -1, \dots$ These are precisely the a_p of [37.a1](#) from Lecture 3 – the curve we used for BSD verification. The second newform corresponds to a different isogeny class, [37.b](#).

Hecke Operators

The **Hecke operators** T_n act on $S_k(\Gamma_0(N))$ as a commuting family of linear endomorphisms. They are diagonalizable (for $(n, N) = 1$).

Definition

A **Hecke eigenform** is a simultaneous eigenvector for all T_n . The eigenvalue of T_p on f equals the p -th Fourier coefficient $a_p(f)$.

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1 > T2 := HeckeOperator(S, 2);
2 > Eigenvalues(T2);      // (eigenvalue, multiplicity)
3 { <0, 1>, <-2, 1> }
4 > T3 := HeckeOperator(S, 3);
5 > Eigenvalues(T3);
6 { <1, 1>, <-3, 1> }
7 > // Each eigenform's a_p = the corresponding eigenvalue of T_p.
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Why this is deep. The same number -2 arises three ways for **37.a1**:

$|E(\mathbb{F}_2)| = 2 + 1 - (-2) = 5$; eigenvalue of T_2 on f ; coefficient of q^2 in the q -expansion.
These agreements *define* modularity.

The Modularity Bridge

The Modularity Theorem

Theorem (Wiles 1995; Taylor–Wiles; Breuil–Conrad–Diamond–Taylor 2001)

Every elliptic curve $E/\mathbb{Q}$ of conductor N corresponds to a weight-2 newform $f \in S_2(\Gamma_0(N))$:

$$L(E, s) = L(f, s).$$

Equivalently, $a_p(E) = a_p(f)$ for every prime p .

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What's hidden in this equality.

- LHS: $a_p(E)$ counts points: $|E(\mathbb{F}_p)| = p + 1 - a_p(E)$. *Geometric.*
- RHS: $a_p(f)$ is a Fourier coefficient. *Analytic.*

Modularity is a wormhole between these worlds.

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Historical aside. Modularity for semistable curves was Wiles's 1995 proof of Fermat's Last Theorem.

Curve $\rightarrow$ Newform: a Live Verification

Take $E = 37.a1$, our running curve from Lecture 3. Compute a_p from point counts:

```
1 > E := EllipticCurve([0, 0, 1, -1, 0]); // 37.a1
2 > Conductor(E);
3 37
4 > for p in PrimesUpTo(20) do
5 >     ap := TraceOfFrobenius(ChangeRing(E, GF(p)));
6 >     printf "a_%o = %o\n", p, ap;
7 > end for;
8 a_2  = -2
9 a_3  = -3
10 a_5  = -2
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Compare with the q -expansion of the *first* newform two slides back. Every coefficient matches. *This* is what Wiles proved.

Newform → Curve via LMFDB

```
1 sage: from lmfdb import db
2 sage: f = db.mf_newforms.lookup("37.2.a.b")
3 sage: f["dim"], f["weight"], f["level"]
4 (1, 2, 37)
5 sage: # Cross-reference: what elliptic curve(s) realize this newform?
6 sage: f['related_objects']
7 ['EllipticCurve/Q/37/b']
8 sage: list(db.ec_curvedata.search({"lmfdb_iso": "37.b"}, ["lmfdb_label"]))
9 [{'lmfdb_label': '37.b1'}, {'lmfdb_label': '37.b2'}, {'lmfdb_label': '37.b3'}]
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The LMFDB web.

- Newform [37.2.a.b](#) has Hecke field $\mathbb{Q}$, $a_p \in \mathbb{Z}$ – the signature of an elliptic-curve newform.
- Three elliptic curves {37.b1, 37.b2, 37.b3} has this newform – it forms an [isogeny class](#). All have the same L -function.
- A sibling newform [37.2.a.a](#) likewise corresponds to the other isogeny class [37.a](#).
- Click through: newform $\rightarrow$ L -function $\rightarrow$ Galois rep $\rightarrow$ elliptic curves. *One click apart.*

Galois Representations: the Mechanism of Modularity

Both sides of modularity attach a **Galois representation**

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- **From E :** $\rho_{E,\ell}$ acts on the Tate module $T_\ell E = \varprojlim E[\ell^n]$.
- **From f (Deligne 1969–1971):** $\rho_{f,\ell}$ acts on a piece of étale cohomology of a modular curve $X_0(N) = \Gamma_0(N) \backslash \mathcal{H}^*$ – the natural compactification of the quotient.

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The Langlands view. Kronecker–Weber classified 1-dimensional Galois representations $\chi : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \mathbb{C}^\times$ via Dirichlet characters: $\chi(\text{Frob}_p)$ is determined by $p \bmod f$.

A 2-dimensional, non-abelian analog: $\rho : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_2(\mathbb{Q}_\ell)$, with $\text{Tr } \rho(\text{Frob}_p) = a_p(f)$ – governed by Fourier coefficients, not congruences. First success of the **Langlands program**, which conjectures n -dimensional Galois reps $\leftrightarrow$ automorphic forms on GL_n .

Research Frontier: Murmurations of Modular Forms

Lecture 3 recap. Pozdnyakov, He–Lee–Oliver (2022): averaging $a_p(E)$ over elliptic curves in a conductor range, broken down by rank, reveals oscillations.

Murmurations, Revisited

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Theorem (Zubrilina, *Invent. math.*, 2025)

*For weight-2 newforms of squarefree level N , the average of $a_f(p)/\sqrt{p}$, broken down by root number, converges as $N \rightarrow \infty$ to an explicit continuous function – the **murmuration density**.*

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This is the **first proved murmuration result**. The tool: the **Eichler–Selberg trace formula** – the natural generalization of the Lecture 3 a_p -averaging into the world of modular forms.

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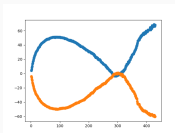
LMFDB note. For trivial-character squarefree-level newforms, the root number $\varepsilon(f)$ is (up to sign convention) the Fricke eigenvalue $w_N(f)$, stored as *fricke_eigenval* on the newform page.

Newform Murmurations in Magma + LMFDB

```
1 sage: from lmfdb import db
2 sage: # Weight 2, level in a range, squarefree, both root numbers
3 sage: query = {"weight": 2, "level": {"$gte": 7500, "$lt": 10000},
4 ...           "level_is_squarefree": True, "char_orbit_label" : "a"}
5 sage: forms = list(db.mf_newforms.search(query,
6 ...           ["fricke_eigenval", "traces"])) # traces = a_n
7 sage: # Group by sign of functional equation
8 sage: forms_by_eps = {1: [], -1: []}
9 sage: for f in forms:
10 ...     forms_by_eps[-f["fricke_eigenval"]].append(f["traces"])
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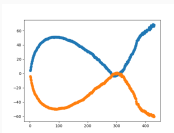
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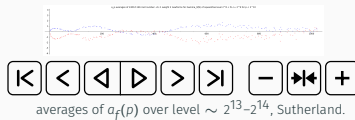
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Newform Murmurations in Magma + LMFDB

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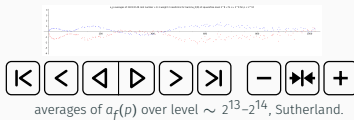
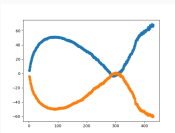
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averages of $a_f(p)$ over level $\sim 2^{13}$ – 2^{14} , Sutherland.

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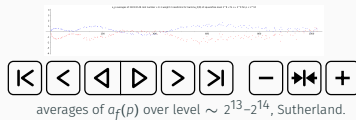
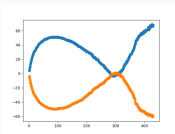


The proved formula. Zubrilina Fig. 3: density $M_8(y)$.

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The L3 to L4 thread. Same phenomenon, different proof. Elliptic-curve murmurations (Lecture 3) are still conjectural; newform murmurations are proved.

Closing the Loop: Theta Series Are Modular Forms

Lecture 1 left us with: for any positive-definite even integral lattice Λ of rank r ,

$$\theta_{\Lambda}(q) = \sum_{v \in \Lambda} q^{Q(v)} = \sum_{m \geq 0} r_{\Lambda}(m) q^m.$$

We computed $\theta_{E_{16}} = \theta_{E_8 \oplus E_8}$. *What kind of object is θ_{Λ} ?*

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And $\theta_{E_{16}} = \theta_{E_8 \oplus E_8}$? Both lie in $M_8(\Gamma_0(1))$, which is also 1-dimensional. Equality is again automatic – modularity explains the Lecture 1 puzzle in one line.

Lattice $\rightarrow$ Newform $\rightarrow$ L-function in Magma + LMFDB

```
1 > // E_8 theta in Magma
2 > L := Lattice("E", 8);
3 > theta<q> := ThetaSeries(L, 10); theta;
4 1 + 240*q^2 + 2160*q^4 + 6720*q^6 + 17520*q^8 + 30240*q^10 + O(q^11)
5 > // = E_4(2*tau): the q^2 substitution from "even" convention
6 > // Strip to natural weight-4 form:
7 sage: f = db.mf_newforms_eis.lookup("1.4.E.a.a")    # level 1, weight 4
8 sage: # Eisenstein E_4 is not a cusp form, so look up in alpha (_eis)
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The LMFDB roundtrip.

1. LMFDB [lattice page](#) for E_8 links to: theta series, Eisenstein series E_4 .
2. E_4 's [page](#) links to the trivial Dirichlet character [1.a](#).
3. The Dirichlet character links to its [L-function](#) $\zeta(s)$. ($L(E, s) = \zeta(s)\zeta(s-3)$)
4. Lecture 1's "can you hear the shape of a drum" failure is, modularly, the statement that two lattices share an L-function – not a coincidence.

Beyond Genus 1: Genus 2 Curves

Genus 2: Why and How

Elliptic curves are genus-1 curves with a rational point. The next case:

Definition

A **genus-2 curve** over $\mathbb{Q}$ has a hyperelliptic model $C : y^2 = f(x)$, with $f \in \mathbb{Q}[x]$ squarefree of degree 5 or 6.

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- No group law on C itself. The group lives on $\text{Jac}(C)$, an abelian surface.
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The LMFDB has genus-2 curves – one of its most distinctive features. Tabulated by conductor up to $\sim 10^6$. Each page links to its Jacobian, its Sato–Tate group, and a modular form associated to its Jacobian (when it exists).

Genus 2 in Magma + LMFDB

```
1 > // Smallest-conductor genus 2 curve over Q: 169.a.169.1
2 > R<x> := PolynomialRing(Rationals());
3 > C := HyperellipticCurve(x^6 + 4*x^5 + 6*x^4 + 2*x^3 + x^2 + 2*x + 1);
4 > Genus(C), Conductor(C);
5 2 169
6 > J := Jacobian(C); #TorsionSubgroup(J);
7 19
8 > Points(C : Bound := 100); // Rational points up to height 100
9 {@ (1 : -1 : 0), (1 : 1 : 0), (-1 : -1 : 1), (-1 : 1 : 1), (0 : -1 : 1), (0 : 1...
10 > RankBounds(J); // in fact J(Q) is finite
11 0 0
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LMFDB tour. Visit [169.a.169.1](#): the curve is linked to the modular form inducing its Jacobian, its L -function, and its Sato–Tate group E_6 .

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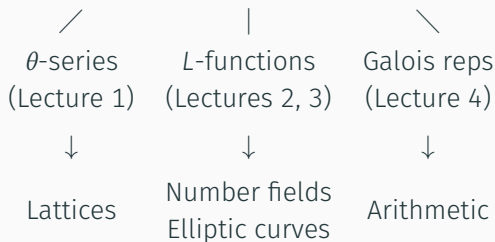
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Research vista. The genus-2 analogue of Mordell–Weil is well understood, but the analogue of BSD – equating rank to order of vanishing of L -function for $\text{Jac}(C)$ – is open.

The Web of the Mini-Course

Modular forms on $\Gamma_0(N)$, weight k



Three big bridges we crossed.

- L1 $\rightarrow$ L4: θ_Λ is a modular form (today).
- L2 $\rightarrow$ L3: H_K via CM elliptic curves (Lecture 3).
- L3 $\rightarrow$ L4: every $E/\mathbb{Q}$ has an attached newform (today).

And one frontier each: genus bound (L1), Cohen–Lenstra (L2), rank-3 BSD + curve murmurations (L3), newform murmurations (L4).

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The LMFDB is the laboratory. Each major discovery in this mini-course (Cohen–Lenstra at scale; murmurations) came from staring at the database. Yours could be next.

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Thank you!

Questions: eranasaf@mit.edu